absolutely no doubt that a half hour earlier the bartender put fully formed ice cubes into the glass; you have no doubt because you trust your memory. And if, by some chance, your confidence regarding what happened during the last half hour should be shaken, you can ask the guy across the way, who was also watching the ice cubes melt (it's a *really* slow night at the bar), or perhaps check the video taken by the bar's surveillance camera, both of which would confirm that your memory is accurate. If you were then to ask yourself what you expect to happen to the ice cubes during the next half hour, you'd probably conclude that they'd continue to melt. And, if you'd gained sufficient familiarity with the concept of entropy, you'd explain your prediction by appealing to the overwhelming likelihood that entropy will increase from what you see, right now at 10:30 p.m., toward the future. All that makes good sense and jibes with our intuition and experience.

But as we've seen, such entropic reasoning—reasoning that simply says things are more likely to be disordered since there are more ways to be disordered, reasoning which is demonstrably powerful at explaining how things unfold toward the future—proclaims that entropy is just as likely to also have been higher in the past. This would mean that the partially melted cubes you see at 10:30 p.m. would actually have been *more* melted at earlier times; it would mean that at 10:00 p.m. they did not begin as solid ice cubes, but, instead, slowly coalesced out of room-temperature water on the way to 10:30 p.m., just as surely as they will slowly melt into room-temperature water on their way to 11:00 p.m.

No doubt, that sounds weird—or perhaps you'd say nutty. To be true, not only would H_2O molecules in a glass of room-temperature water have to coalesce spontaneously into partially formed cubes of ice, but the digital bits in the surveillance camera, as well as the neurons in your brain and those in the brain of the guy across the way, would all need to spontaneously arrange themselves by 10:30 p.m. to attest to there having been a collection of fully formed ice cubes that melted, even though there never was. Yet this bizarre-sounding conclusion is where a faithful application of entropic reasoning—the same reasoning that you embrace without hesitation to explain why the partially melted ice you see at 10:30 p.m. continues to melt toward 11:00 p.m.—leads when applied in the time-symmetric manner dictated by the laws of physics. This is the